Simulation study: mixture of experts model to check identifiability

The Tien Mai & Zhi Zhao

Last updated: 25 November, 2025

Data simulation:

- Sample size $n = 200$
- Dimension of the Wishart distribution $p = 2$
- Number of latent components $K = 2$
- Probabilities of subpopulations $\boldsymbol{\pi} = (0.8, 0.2)$
- Degrees of freedom $\boldsymbol{\nu} = (3, 12)$
- Scale matrices of the Wishart distribution Σ_1, Σ_2
- Data $S_i \sim \pi_1 \text{Wishart}(\nu_1, \Sigma_1) + \pi_2 \text{Wishart}(\nu_2, \Sigma_2)$

Modeling:

- Bayesian mixture of experts of two Wishart distributions

```r
rm(list = ls())

library(mewishart)

set.seed(123)
n <- 200 # number of subjects
p <- 2

dat <- simData(n, p,
  pis = c(0.8, 0.2),
  nus = c(3, 12),
  Sigma = NULL # default Sigmas pre-defined in function 'simData()'
)
S_list <- dat$S
Sigma_list <- dat$Sigma_list

my_K_intial <- 2

# run a mewishart model
set.seed(123)
fit <- mewishart(S_list,
  K = my_K_intial, init_pi = c(0.9, 0.1),
  niter = 10000, burnin = 1000, thin = 1, verbose = TRUE
)
```

```
Iter    1 | LL=-492.5 | 709.2 iter/sec | acc_rate_nu=[0.000 0.000]
Iter  500 | LL=-452.0 | 907.6 iter/sec | acc_rate_nu=[0.050 0.064]
Iter 1000 | LL=-453.3 | 953.0 iter/sec | acc_rate_nu=[0.048 0.061]
Iter 1500 | LL=-457.4 | 968.4 iter/sec | acc_rate_nu=[0.047 0.063]
Iter 2000 | LL=-452.7 | 974.8 iter/sec | acc_rate_nu=[0.046 0.065]
Iter 2500 | LL=-452.9 | 973.9 iter/sec | acc_rate_nu=[0.046 0.066]
```

```
Iter 3000 | LL=-456.8 | 979.9 iter/sec | acc_rate_nu=[0.047 0.067]
Iter 3500 | LL=-452.9 | 974.0 iter/sec | acc_rate_nu=[0.046 0.067]
Iter 4000 | LL=-451.8 | 975.8 iter/sec | acc_rate_nu=[0.044 0.067]
Iter 4500 | LL=-452.5 | 980.1 iter/sec | acc_rate_nu=[0.043 0.065]
Iter 5000 | LL=-452.2 | 982.9 iter/sec | acc_rate_nu=[0.043 0.062]
Iter 5500 | LL=-452.0 | 984.7 iter/sec | acc_rate_nu=[0.042 0.060]
Iter 6000 | LL=-454.2 | 988.0 iter/sec | acc_rate_nu=[0.042 0.059]
Iter 6500 | LL=-454.3 | 990.7 iter/sec | acc_rate_nu=[0.041 0.059]
Iter 7000 | LL=-455.6 | 991.9 iter/sec | acc_rate_nu=[0.042 0.059]
Iter 7500 | LL=-454.0 | 991.5 iter/sec | acc_rate_nu=[0.041 0.059]
Iter 8000 | LL=-451.0 | 990.0 iter/sec | acc_rate_nu=[0.041 0.060]
Iter 8500 | LL=-453.6 | 990.8 iter/sec | acc_rate_nu=[0.041 0.060]
Iter 9000 | LL=-452.8 | 992.3 iter/sec | acc_rate_nu=[0.041 0.061]
Iter 9500 | LL=-453.5 | 993.3 iter/sec | acc_rate_nu=[0.041 0.061]
Iter 10000 | LL=-453.1 | 994.1 iter/sec | acc_rate_nu=[0.040 0.061]
```

```r
# posterior summaries
colMeans(fit$pi)
```

```
[1] 0.7891114 0.2108886
```

```r
colMeans(dat$pi) # true pi
```

```
[1] 0.8 0.2
```

```r
apply(fit$nu, 2, mean)
```

```
[1]  3.035037 10.195513
```

```r
dat$nu # true nu
```

```
[1]  3 12
```

Check identifiability issue

```r
# run the model again with swapped initial weights
set.seed(1234)
fit2 <- mewishart(S_list,
  K = my_K_intial, init_pi = c(0.1, 0.9),
  niter = 10000, burnin = 1000, thin = 1, verbose = TRUE
)
```

```
Iter    1 | LL=-492.5 | 1019.5 iter/sec | acc_rate_nu=[0.000 0.000]
Iter  500 | LL=-451.0 | 995.0 iter/sec | acc_rate_nu=[0.040 0.060]
Iter 1000 | LL=-453.9 | 994.7 iter/sec | acc_rate_nu=[0.038 0.056]
Iter 1500 | LL=-455.2 | 1002.2 iter/sec | acc_rate_nu=[0.038 0.057]
Iter 2000 | LL=-452.9 | 999.2 iter/sec | acc_rate_nu=[0.041 0.052]
Iter 2500 | LL=-456.3 | 1002.4 iter/sec | acc_rate_nu=[0.042 0.054]
Iter 3000 | LL=-453.9 | 1001.6 iter/sec | acc_rate_nu=[0.042 0.053]
Iter 3500 | LL=-458.2 | 999.2 iter/sec | acc_rate_nu=[0.041 0.053]
Iter 4000 | LL=-453.8 | 996.5 iter/sec | acc_rate_nu=[0.041 0.056]
Iter 4500 | LL=-452.8 | 989.0 iter/sec | acc_rate_nu=[0.042 0.056]
Iter 5000 | LL=-453.4 | 987.6 iter/sec | acc_rate_nu=[0.042 0.059]
Iter 5500 | LL=-452.1 | 983.6 iter/sec | acc_rate_nu=[0.044 0.057]
Iter 6000 | LL=-453.3 | 978.7 iter/sec | acc_rate_nu=[0.043 0.058]
Iter 6500 | LL=-453.2 | 979.8 iter/sec | acc_rate_nu=[0.042 0.056]
Iter 7000 | LL=-456.5 | 982.1 iter/sec | acc_rate_nu=[0.043 0.055]
```

```
Iter 7500 | LL=-452.7 | 980.1 iter/sec | acc_rate_nu=[0.044 0.056]
Iter 8000 | LL=-453.0 | 980.7 iter/sec | acc_rate_nu=[0.043 0.056]
Iter 8500 | LL=-456.8 | 981.6 iter/sec | acc_rate_nu=[0.043 0.056]
Iter 9000 | LL=-456.1 | 983.2 iter/sec | acc_rate_nu=[0.043 0.057]
Iter 9500 | LL=-452.0 | 984.2 iter/sec | acc_rate_nu=[0.042 0.057]
Iter 10000 | LL=-452.0 | 983.5 iter/sec | acc_rate_nu=[0.043 0.057]
```

```r
# posterior summaries
colMeans(fit2$pi)
```

```
[1] 0.2090747 0.7909253
```

```r
colMeans(dat$pi) # true pi
```

```
[1] 0.8 0.2
```

```r
apply(fit2$nu, 2, mean)
```

```
[1] 10.643540  3.012158
```

```r
dat$nu # true nu
```

```
[1]  3 12
```

Conclusion:

The two model fittings above with initial weights `init_pi=c(0.9, 0.1)` and `init_pi=c(0.1, 0.9)` result in similar results, which indicate possibly identifiability issue.

It might be NOT possible based on the proof of Proposition 2 in Yakowitz and Spragins (1968) to give similar proofs to show identifiability of the MoE of (inverse) Wishart distributions.